Ex. 7. *If the sum of two numbers is 10 and the sum of their reciprocals is* $\frac{5}{12}$*, find the numbers.* **(P.C.S., 2006)**

Sol. Let the numbers be x and y.

Then, $x + y = 10$ ...(*i*)

And, $\frac{1}{x}+\frac{1}{y}=\frac{5}{12} \Rightarrow \frac{x+y}{xy} = \frac{5}{12} \Rightarrow xy = \frac{10\times 12}{5} = 24$ ...(*ii*)

$\therefore \quad x - y = \sqrt{(x+y)^2 - 4xy} = \sqrt{(10)^2 - 4\times 24} = \sqrt{100-96} = \sqrt{4} = 2 \Rightarrow x - y = 2$ ...(*iii*)

Adding (*i*) and (*iii*), we get: $2x = 12$ or $x = 6$.

Putting $x = 6$ in (*i*), we get: $y = 4$.

Hence, the required numbers are 6 and 4.

Ex. 8. *Three numbers are in the ratio 3 : 2 : 5. The sum of their squares is 1862. Find the numbers.* **(R.R.B., 2007)**

Sol. Let the numbers be $3x$, $2x$ and $5x$.

Then, $(3x)^2 + (2x)^2 + (5x)^2 = 1862 \Rightarrow 9x^2 + 4x^2 + 25x^2 = 1862$

$\Rightarrow 38x^2 = 1862 \Rightarrow x^2 = \frac{1862}{38} = 49 \Rightarrow x = \sqrt{49} = 7.$

Hence, the numbers are 21, 14 and 35.

Ex. 9. *The sum of seven consecutive natural numbers is 1617. How many of these numbers are prime?* **(S.S.C., 2006)**

Sol. Let the seven consecutive numbers be x, $(x + 1)$, $(x + 2)$, $(x + 3)$, $(x + 4)$, $(x + 5)$ and $(x + 6)$.

Then, $x + (x + 1) + (x + 2) + (x + 3) + (x + 4) + (x + 5) + (x + 6) = 1617$

$\Rightarrow 7x + 21 = 1617 \Rightarrow 7x = 1596 \Rightarrow x = 228.$

Thus, the numbers are 228, 229, 230, 231, 232, 233 and 234.

Of these numbers, only two numbers i.e. 229 and 233, are prime.

Ex. 10. *The product of two consecutive numbers is 4032. Find the numbers.* **(Bank P.O., 2008)**

Sol. Let the numbers be x and $(x + 1)$.

Then, $x(x + 1) = 4032 \Rightarrow x^2 + x - 4032 = 0 \Rightarrow x^2 + 64x - 63x - 4032 = 0$

$\Rightarrow x(x + 64) - 63(x + 64) = 0 \Rightarrow (x + 64)(x - 63) = 0$

$\Rightarrow x = 63.$ $\quad [\because x \neq -64]$

Hence, the required numbers are 63 and 64.

Ex. 11. *The sum of two numbers is 15 and the sum of their squares is 113. Find the numbers.* **(R.R.B., 2006)**

Sol. Let the numbers be x and $(15 - x)$.

Then, $x^2 + (15 - x)^2 = 113 \Rightarrow x^2 + 225 + x^2 - 30x = 113$

$\Rightarrow 2x^2 - 30x + 112 = 0 \Rightarrow x^2 - 15x + 56 = 0$

$\Rightarrow (x - 7)(x - 8) = 0 \Rightarrow x = 7$ or $x = 8$.

So, the numbers are 7 and 8.

Ex. 12. *The average of four consecutive even numbers is 27. Find the largest of these numbers.*

Sol. Let the four consecutive even numbers be x, $x + 2$, $x + 4$ and $x + 6$.

Then, sum of these numbers $= (27 \times 4) = 108$.

So, $x + (x + 2) + (x + 4) + (x + 6) = 108$ or $4x = 96$ or $x = 24$.

$\therefore$ Largest number $= (x + 6) = 30$.

Ex. 13. *The sum of the squares of three consecutive odd numbers is 2531. Find the numbers.* **(R.R.B., 2010)**

Sol. Let the numbers be x, $x + 2$ and $x + 4$.

Then, $x^2 + (x + 2)^2 + (x + 4)^2 = 2531 \Rightarrow 3x^2 + 12x - 2511 = 0 \Rightarrow x^2 + 4x - 837 = 0$

$\Rightarrow (x - 27)(x + 31) = 0 \Rightarrow x = 27.$

Hence, the required numbers are 27, 29 and 31.